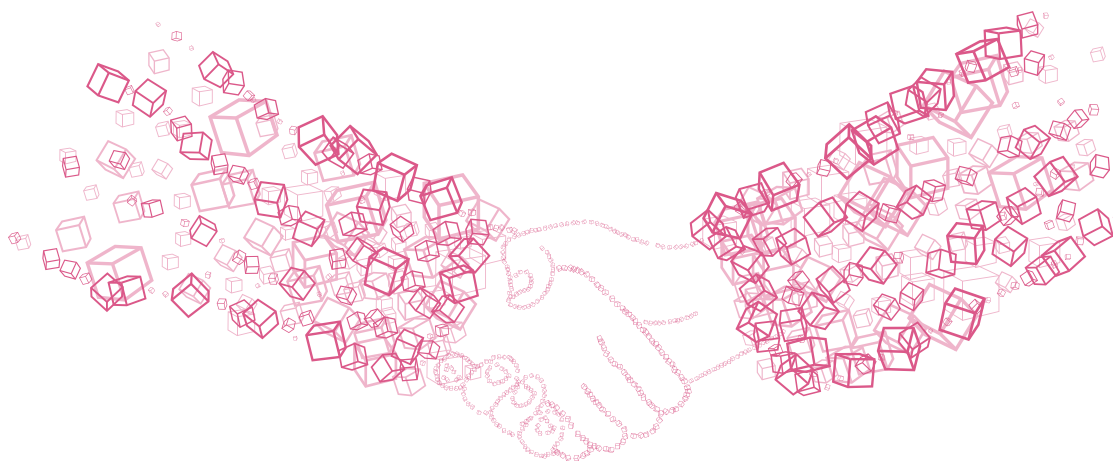


# DAML: The Programming Language for Smart Contracts in a Blockchain



The DAML smart contract language is a purpose-built domain specific language designed to encode the shared business logic of an application. It is used for the development and deployment of distributed applications in the blockchain environment.

**B**lockchain technology is a secure mechanism to keep track of information in a way that makes it hard or impossible to modify or hack it. A blockchain integrates the digital ledger of transactions, which is copied and sent to every computer on its network. In each block of the chain, there are a number of transactions. When a new transaction takes place on the blockchain, a record of that transaction is added to the ledgers of everyone who is part of the chain.

Blockchain uses distributed ledger technology (DLT), in which a database isn't kept in one server or node. In a blockchain, transactions are recorded with an immutable cryptographic sign known as a hash. This means that if one block in one channel or chain is changed, it will be hard for hackers to change that block in the chain, as

they would have to do this for every single version of the chain that is out there. Blockchains, like Bitcoin and Ethereum, keep growing as new blocks are added to the chain, which makes the ledger safer.

With the implementation of smart contracts in blockchain, there is automatic execution of scenarios without any human intervention. Smart contract technology makes it possible to enforce the highest level of security, privacy and anti-hacking implementations.

The use cases and applications of blockchain are:

- Cryptocurrencies
- Smart contracts
- Secured personal information
- Digital health records
- E-governance
- Non-fungible tokens (NFTs)

- Gaming
- Cross-border financial transactions
- Digital voting
- Supply chain management

As per *Statista.com*, the size of the blockchain technology market is increasing at a very fast speed since the last few years and is predicted to touch US\$ 40 billion by 2025.

## Programming languages and toolkits for blockchain

A number of programming languages and development toolkits are available for distributed applications and smart contracts. Programming and scripting languages for the blockchain include Solidity, Java, Vyper, Serpent, Python, JavaScript, GoLang, PHP, C++, Ruby, Rust, Erlang, etc, and are employed depending upon the implementation scenarios and use cases.